Jet Joe Lab

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"All members contributed equally and collaborated on each deliverable."
Signed,
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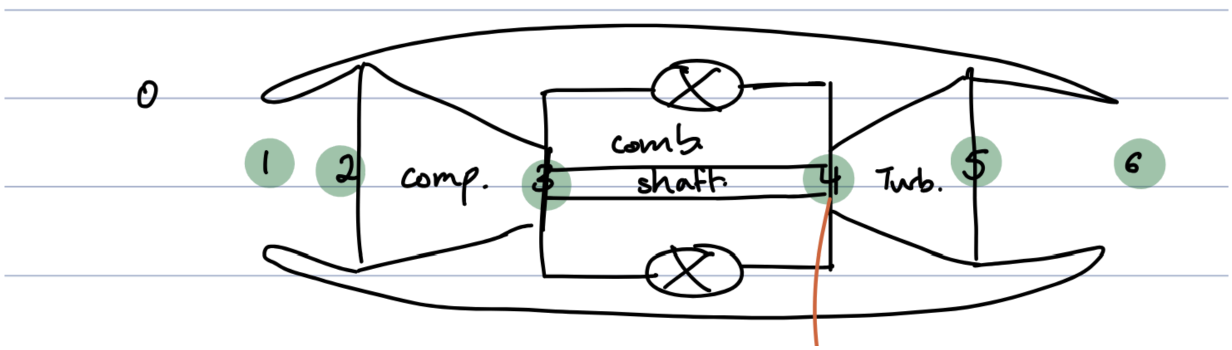
Deliverable A

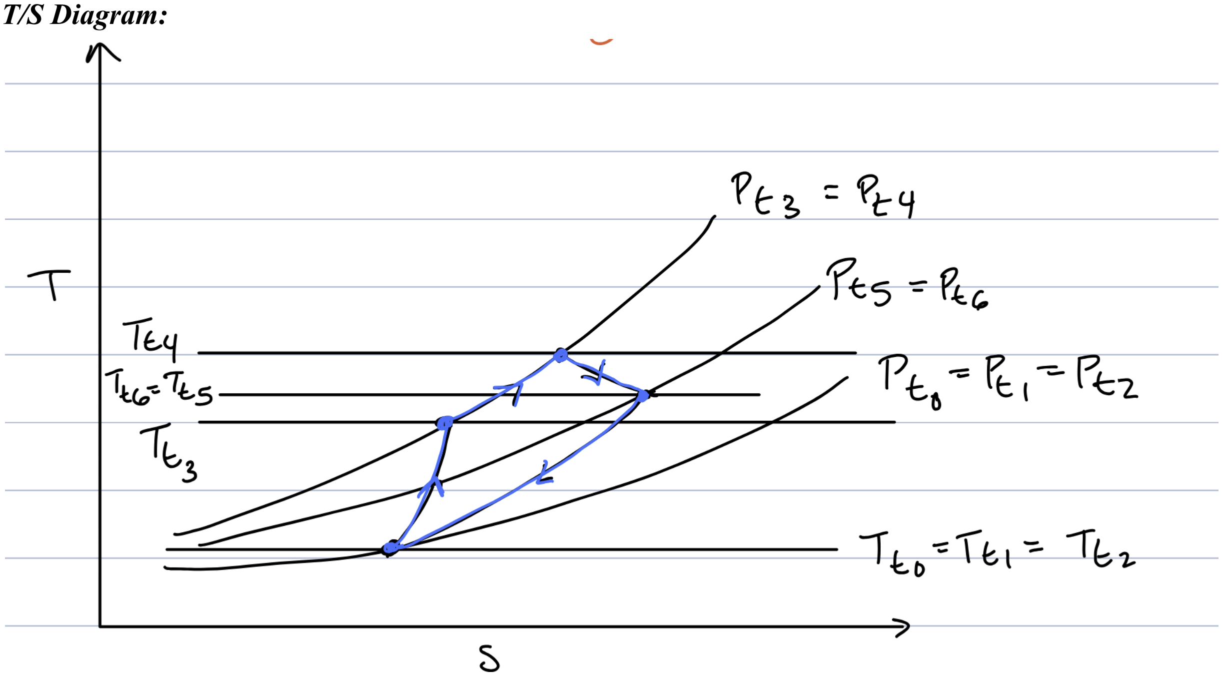
Diagram of JJ Engine with stations 0-6:



Assumptions:

1. Compressor and Turbine are adiabatic
2. Choked flow happens at the Nozzle Guide Veins
3. EGT= T_{t5} because the flow stagnates on the temperature probe
4. Adiabatic turbine and compressor efficiency= 70%
5. H_f= 43.15 mJ/Kg
6. Combustor has a 100% efficiency
7. Part 3 assumes that $\dot{m}_f$ is negligible when solving for Π_c since the deliverable states to use specific shaft work
8. Inlet and outlet nozzle have adiabatic reversible flow

T/S Diagram:



Mean Line Model:
Inputs:

1. Inner diameter of the turbine (in)
2. Outer diameter of the turbine (in)
3. Depth of the stator blade (used to calculate area normal to the flow through the stator blades) (in)
4. Number of stator blades
5. Beta 1 (turbine rotor inlet angle) (degrees)
6. Omega (RPM)

Outputs:

1. Specific fuel consumption (kg/s/N)
2. Specific thrust (N/kg/s)
3. T_{t4} (Kelvin)
4. Inlet Mass flow (kg/s)
5. Cycle pressure ratio

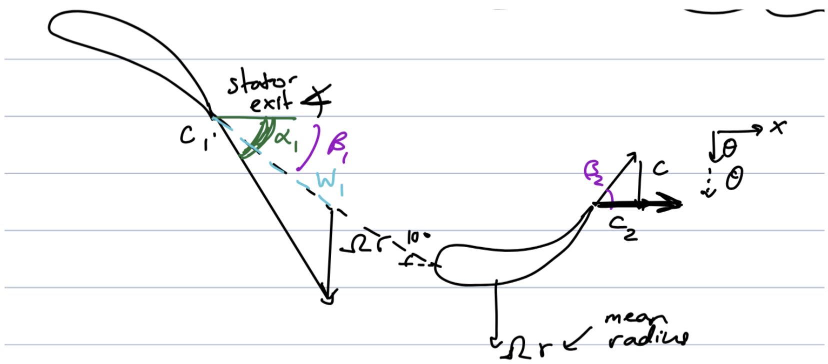
Constants:

1. EGT (Kelvin)
2. P_{inf} (atmospheric pressure) (Pa)
3. T_{inf} (atmospheric temperature) (Kelvin)
4. Mach number at choked point is 1
5. γ_T (gamma of turbine)
6. γ_C (gamma of compressor)
7. R_T (R of turbine)
8. R_C (R of compressor)

Comments can be found throughout the code discussing how we structured our model using functions for each of the equations we found and variables to hold both inputs and calculated outputs. A breakdown of our overall process can be found below. The code has been attached as separate .py files.

Deliverables Solution Process:

1) Determine turbine specific shaft work using Euler turbomachinery equation, non-dimensional velocity triangles and turbine blade angle measurements (NGV)



Based on the velocity triangles, we were able to find $C_{\theta 1}$ in terms of the relative velocity W_1, the angles α_1, β_1 and the Ωr where r is the mean radius of the turbine rotor. Through algebraic manipulation, we solved for $C_{\theta 1}$ in terms of known angles and measured geometric quantities. Note that the angle α_1 was calculated using measured properties of the stator, including inner and outer radii, number of blades, and the area normal to the flow. See the appendix for the derivations.

Using the Euler turbomachinery equation, the turbine specific shaft work was calculated.

2) Assume adiabatic turbine and compressor efficiencies

Adiabatic turbine efficiency = 70%

Adiabatic compressor efficiency = 70%

These values are estimated based on a research paper "Very Small Gas Turbine Jet Engines - Current Limits and Potential for Improvement" by Hirndorf et. al. discussing turbines and compressors of similar scale (on order of magnitude).

3) Determine compressor pressure ratio using specific shaft work balance

Because the compressor and turbine operate the same shaft, the specific turbine shaft work must equal the specific compressor shaft work. The specific compressor shaft work can be written in terms of the ideal specific compressor shaft work and the adiabatic compressor efficiency. The ideal specific compressor shaft work is a function of the compressor pressure ratio. This equation was then set equal to the work done by the turbine w^T_{actual} previously solved with the fuel fraction ($\dot{m}_f / \dot{m}_c$) and the pressure ratio of the combustor as unknowns.

However we assume that $\dot{m}_f$ is negligible when solving for Π_c since the deliverable states to use specific shaft work; the fuel fraction cancels out and the compressor pressure ratio can be solved for.

4) Given EGT and fuel heating value determine turbine inlet temperature using shaft work balance and heat release

Using a control volume around the turbine the enthalpy of the turbine can be written in terms of cp^T, T_{t4}, and T_{t5} with $T_{t5} = T_{t6}$ = EGT. This is because we assume that the flow is stagnated on the temperature probe. We solve for the work coefficient λ using known/measured geometric quantities. We can use the λ value to solve for T_{t4}.

Looking at a control volume around the combustor, we set up a first law problem considering the addition of fuel mass flow; the work of the turbine and compressor cancel out. We have two unknowns in the resulting equation: f (fraction of fuel mass flow to air mass flow) and $T_{t3.}$
The actual shaft work in the turbine was set to equal the actual shaft work in the compressor to find an additional equation with the two unknowns f and T_{t3}. From these equations, the fuel mass fraction was solved numerically.

5) Assuming choked NGVs and measured turbine area, determine turbine mass flow
Using the mass flow equation provided in the slides, $\dot{m}_4^T$ was solved for. This equation is the mass flow at the choked point in the NGV so that Mach number could be assumed to be 1. Stagnation pressure and temperature are assumed to be conserved between station 4 and the NGVs such that $P_t = P_{t4}$, $T_t = T_{t4}$ and the measured turbine area is A_4.

6) Determine exhaust nozzle kinetic energy
The pressure ratio across the turbine was first found using 1st law of the turbine and multiplying the actuarial turbine work by the adiabatic turbine efficiency with T_{t4} = EGT . This pressure ratio between station 4 and 5 was then used to find the temperature at station 6 which was used to find the speed of sound with $a=\sqrt{\gamma RT}$. Assuming that the flow is adiabatic reversible the Mach number at 6 was found. Speed was found as a function of Mach number and speed of sound at station 6. Nozzle kinetic energy was solved using $KE = mc^2/2$, with m representing the mass flow of the air and fuel out of the engine.

7) Determine specific thrust and specific fuel consumption
Thrust is equal to the mass times the exit speed, so the specific thrust is equal to the exhaust exit velocity. The units of speed are m/s as expected (since if we multiply by specific mass flow units kg/s we get units of force).
Specific fuel consumption was calculated using the fuel fraction and exhaust exit speed.
Finally, exhaust exit speed was used to calculate the flow coefficient.

8) Finding cycle pressure ratio
Cycle pressure ratio is the ratio of the highest to lowest stagnation pressures so
Cycle pressure ratio = p_{t4} / p_{t0}

Governing equations, and final expressions for the performance metric:
1) Determine turbine specific shaft work using Euler turbomachinery equation, non-dimensional velocity triangles and turbine blade angle measurements (NGV)

$$c_{\theta_1} = \frac{\Omega \bar{r} \tan(\alpha_1)}{\tan(\alpha_1) - \tan(\beta_1)}$$

$$\alpha_1 = \cos^{-1}\left(\frac{(r_2 - r_1) \times N_{\text{blades}} \times \text{Depth measured}}{\pi (r_2^2 - r_1^2)} \right)$$

$$W^T_{\text{actual}} = h_{t4} - h_{t5} = \Omega (r_1 c_{\theta 1} - r_2 c_{\theta 2})$$

3) Determine compressor pressure ratio using specific shaft work balance

$$\pi_c = \left(\frac{\eta^c W^T_{\text{actual}}}{T_{t2}\, c_p^c} + 1 \right)^{\frac{\gamma^c}{\gamma^c - 1}} \quad \text{for} \quad \pi_c = \frac{P_{t3}}{P_{t2}} \quad \& \quad T_{t2} = T_{t1} = T_\infty$$

4) Given EGT and fuel heating value determine turbine inlet temperature using shaft work balance and heat release

$$\frac{C_{\theta 1} \Omega \bar{r}}{c_p} + EGT = T_{t4}$$

$$T_{t3} = \frac{(1+f)\, c_p^T (T_{t4} - T_{t5})}{c_p^c} + T_{t2}$$

$$\frac{\frac{f h_f}{1+f} + T_{t4}\, c_p^T}{c_p^c} = T_{t3}$$

5) Assuming choked NGVs and measured turbine area, determine turbine mass flow

$$\dot{m}_4^T = \frac{M A_4 P_{t4} \sqrt{\gamma_T}}{\sqrt{R^T T_{t4}} \left(1 + \frac{\gamma_T - 1}{2} M^2 \right)^{1/2 \left(\frac{\gamma_T + 1}{\gamma_T - 1} \right)}}, \quad M = 1 \text{ choked}$$

$$\dot{m}_1 = \frac{\dot{m}_4^T}{1+f}$$

6) Determine exhaust nozzle kinetic energy

$$\pi^T = \left(1 + \frac{W^T_{\text{actual}}}{c_p^T T_{t5} \eta^T_{ad}} \right)^{\frac{\gamma^T}{1-\gamma^T}}$$

where $T_{t5} = EGT$

$$P_{t5} = \pi^T P_{t4} = \pi^T P_{t3} \qquad P_{t5} = P_{t6}$$

$$M_6 = \sqrt{\frac{2}{\gamma^T - 1} \left(\left(\frac{P_{t5}}{P_\infty} \right)^{\frac{\gamma^T - 1}{\gamma^T}} - 1 \right)}$$

$$T_6 = T_{t6} \frac{1}{1 + \frac{\gamma^T - 1}{2} M_6^2} \qquad T_{t6} = T_{t5}$$

$$a_6^T = \sqrt{\gamma^T R^T T_6}$$

$$C_6 = a^T \left[\frac{2}{\gamma^T - 1} \left[\left(\frac{P_{t5}}{P_\infty} \right)^{\frac{\gamma^T - 1}{\gamma^T}} - 1 \right] \right]^{\frac{1}{2}}$$

$$KE_6 = \frac{\dot{m}_c (1+f) C_6^2}{2}$$

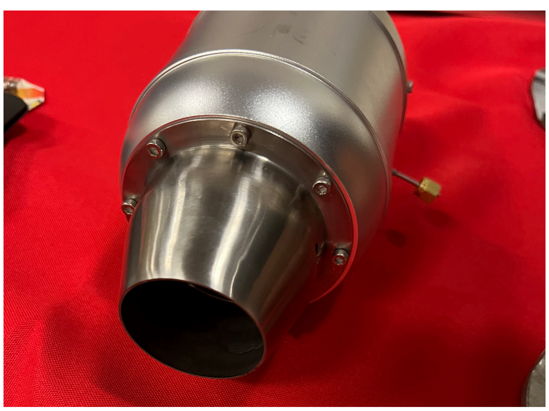
7) Determine specific thrust and specific fuel consumption

$$T = \text{specific thrust} = \frac{F}{\dot{m}_1 a_0} = \frac{a_6 M_6 \dot{m}_4^T}{\dot{m}_1 a_0} \qquad \frac{c_6}{\Omega r} = \phi$$

$$TSFC = \frac{f}{c_6 (1+f)}$$

Deliverable B

Assembled Engine:



Coast-Down Test:

1. The engine was mounted on the test stand, and the starter motor and cowling were attached to the front. The glow plug was connected (though not used), and Joe double-checked that the fuel lines were disconnected (to avoid an accidental start).
2. An RPM sensor was connected to an output monitor, so we could view the RPM of the engine during the test.

3. Joe started the start-up sequence, and the starter motor spun up the engine as much as it could. The target was ~4000 rpm, however our engine only reached ~900. This is due to the engine never having been operated prior to this test.
4. We measured the spin down time to be 0.66 seconds. Again, this was faster than desired. However, Joe checked that our engine was assembled correctly, ensuring that no part was rubbing on the casing. We determined that since the engine had not been fired, oil had been unable to work its way through the system to lubricate it, and thermal cycles had not caused the parts to 'settle', both of which affected performance even though the engine was assembled correctly.

Measured Parameters:

turbine rotor inlet angle	31	deg
depth of stator outlet	0.225	in
number of stator blades	13	#
outer diameter of the rotor	2.165	in
inner diameter of the rotor	1.442	in
exit nozzle diameter	1.641	in

Deliverable C

Quantity	**Value**	**Units**
Cycle Pressure Ratio:	2.034	unitless
Inlet Mass Flow:	0.1656	kg/s
Turbine Inlet Stagnation Temperature:	1047	K
Work Coefficient:	0.7897	unitless
Specific Thrust:	1.093	unitless
Specific Fuel Consumption:	5.485e-05	(kg/s)/N
Specific Fuel Consumption:	1.937	(lbm/hr)/lbf

Justification and assumptions:
We assume $\gamma_C = 1.4$ since air is the working fluid, while $\gamma_T = 1.3$ is the specific heat ratio considering the combustion fluid. All other constants (specific gas constant, atmospheric pressure, and atmospheric temperature, EGT) are known/searched up values.

The results of the model depend on the assumed adiabatic efficiencies of the compressor and turbine. We assumed these efficiencies to be 70%, based on a research paper "Very Small Gas Turbine Jet Engines - Current Limits and Potential for Improvement" by Hirndorf et. al. discussing turbines and compressors of similar scale (on order of magnitude). This paper reported an efficiency of 73% but was slightly larger than our engine justifying our choice of 70% efficiency.

FHV was assumed to be 43.15 MJ/kg based on the researched value of Jet A1.

Our calculated values align with the expected values.
Specific thrust converted to actual thrust gets us to around 13.83 lbs of thrust force which is close to the expected thrust force of 14 lbs. Actual fuel consumption was expected to be 6-7 oz per min and calculated to be 7.15 oz per min. Fuel fraction was expected to be 50:1 (2%) based on research and found to be 2.04%.

Deliverable D

Thermodynamic Assessment

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FOR THERMODYNAMIC ASSESSMENT
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Collected Data:
             RPMs[rpm]    P0[psia]     P0-P1[in-h20]  T0[°C]       P_t3[psia]   T_t3[°C]     F[lbf]       mdot_f[gal/hr]  EGT[°C]
Pre-Test     0            14.82        0.001          21.3         14.6         47.9         0.1          0.0             84
Idle Speed   45000        14.82        0.12           21.3         16.0         35.9         0.9          0.74            588
Speed 1      72700        14.82        0.34           21.6         19.3         43.7         2.3          0.91            524
Speed 2      135600       14.82        1.515          22.6         30.0         92.8         10.0         2.77            610
Max Speed    160000       14.82        2.06           23.3         37.0         128.4        14.8         4.24            772
Post-Test    0            14.82        0.001          22.4         14.6         38.1         0.3          0.0             116
------------------------------------
Measured Data:
             P0[Pa]       T_t1[K]      P_t3[Pa]       T_t3[K]      T_t5[K]      f[-]
Pre-Test     102180.936   294.45       100664.08      321.05       357.15       0.0
Idle Speed   102180.936   294.45       110316.8       309.05       861.15       0.012
Speed 1      102180.936   294.75       133069.64      316.85       797.15       0.008
Speed 2      102180.936   295.75       206844.0       365.95       883.15       0.013
Max Speed    102180.936   296.45       255107.6       401.55       1045.15      0.017
Post-Test    102180.936   295.55       100664.08      311.25       389.15       0.0
------------------------------------
Estimated Parameters:
             n_c [-]      T_t4 [K]     T_t5S [K]      c_6 [m/s]    Specific Thrust [-]
Pre-Test     1.0          378.635      833.809        0.0          0.0
Idle Speed   0.446        872.801      846.696        0.0          0.0
Speed 1      1.045        814.864      797.92         278.621      0.816
Speed 2      0.94         939.134      879.604        451.2        1.326
Max Speed    0.843        1128.635     1029.567       540.629      1.593
Post-Test    1.0          401.831      559.808        0.0          0.0
```

Adiabatic Efficiency:

- Set to 1 for Pre and Post-Test, since there is no work done and the efficiency expression evaluates to 0/0.
- For Idle speed, the efficiency is probably artificially low, since the compressor shell would still have been hot from pre-heating.
- For other speeds, the efficiency is certainly artificially high (it is impossible for actual efficiency to be greater than 1), since the compressor shell would not have been responsive enough to temperature changes to rise to the actual air temperature in the few seconds that each speed was held.

Isentropic Turbine Exit Temp:

- This should be equal to actual turbine exit temp for Pre and Post-Test, but because the engine was pre-heated, and remains hot after max speed, these numbers are calculated as increased from actual T_{t5}.

Exit Velocity and Specific Thrust:

- Manually set to 0 for Idle, Pre, and Post-Test, since preheating/residual heat messed up the calculation, and actually put a negative number (whose magnitude was near 0) in the square root for Idle speed.

Direct Assessment

FOR DIRECT ASSESSMENT

Collected Data:

	RPMs[rpm]	P0[psia]	P0-P1[in-h20]	T0[°C]	P_t3[psia]	T_t3[°C])	F[lbf]	mdot_f[gal/hr]	EGT[°C]
Pre-Test	0	14.82	0.001	21.3	14.6	47.9	0.1	0.0	84
Idle Speed	45000	14.82	0.12	21.3	16.0	35.9	0.9	0.74	588
Speed 1	72700	14.82	0.34	21.6	19.3	43.7	2.3	0.91	524
Speed 2	135600	14.82	1.515	22.6	30.0	92.8	10.0	2.77	610
Max Speed	160000	14.82	2.06	23.3	37.0	128.4	14.8	4.24	772
Post-Test	0	14.82	0.001	22.4	14.6	38.1	0.3	0.0	116

Measurement Data

	Cell Presure [Pa]	Inlet Temperatures[K]	Thrust[N]	Inlet Area [m^2]	Inlet Pressure [Pa]
Pre-Test	102180.936	294.45	0.445	0.006	102180.687
Idle Speed	102180.936	294.45	4.003	0.006	102151.044
Speed 1	102180.936	294.75	10.231	0.006	102096.242
Speed 2	102180.936	295.75	44.482	0.006	101803.55
Max Speed	102180.936	296.45	65.833	0.006	101667.79
Post-Test	102180.936	295.55	1.334	0.006	102180.687

Estimated Params

	M1[-]	Mass flow [kg/s]	Specific Thrust [-]	SFC [(lbf/hr)/lbf]
Pre-Test	0.002	0.005	0.268	0.0
Idle Speed	0.02	0.053	0.221	5.626
Speed 1	0.034	0.089	0.335	2.707
Speed 2	0.073	0.187	0.691	1.896
Max Speed	0.085	0.217	0.878	1.96
Post-Test	0.002	0.005	0.805	0.0

Comparison of Assessments with Model

	Press. Ratio (-)	Inlet Mass Flow (kg/s)	Turbine Inlet Temp. (K)	Spec. Thrust (-)	Spec. Fuel Consump. (lbm/hr)/lbf
Model	2.034	0.168	1046.672	1.093	1.927
Thermo Assessment	-	-	1128.635	1.593	-
Direct Assessment	2.497	0.217	-	0.878	1.96
% Diff w/ Model & Direct	18.529	22.807	-	24.478	1.716
% Diff w/ Model & Thermo	-	-	7.262	31.409	-

Note that the raw collected data is displayed at the top of the results for both the thermodynamic assessment and the direct assessment.

Our model predicted a pressure ratio which was notably less than the direct assessment. This is likely due to our assumed efficiency—70%—being greater than the actual efficiency. If we had assumed a lower efficiency, we would have predicted a higher pressure ratio in our model, since a less efficient compressor must operate at a higher pressure ratio to perform the same amount of work. Additionally, we predicted slightly less thrust, and thus less compressor work.

The model predicted an inlet mass flow that was 23% lower than the measured inlet mass flow in the direct assessment. In our model, we calculated the inlet mass flow based on the mass flow at the turbine inlet. We assumed that Mach number was 1 (choked flow) at this station. However, we may have mismeasured/miscalculated the choked area. In order to find the area at this station, we had to estimate the depth of the NGVs using a caliper, which is error prone. If the calculated area was smaller than the true choked area, that may have contributed to the lower estimated inlet mass flow.

The model predicted a turbine inlet temperature that was 7.3% lower than the temperature predicted by the thermodynamic assessment. The two variables which could have raised the thermodynamic prediction were compressor skin temp and EGT. During the run, EGT fluctuated by a few degrees; measuring at the high point of this fluctuation could increase the predicted turbine inlet temp. Similarly, the compressor casing may have received excess heat from the combustor region, artificially raising that temperature, and thus raising the predicted inlet temperature.

Compared to our model, the direct assessment of specific thrust is fairly similar, but somewhat smaller. This is probably because our model assumes 70% efficiency, and the real turbine is slightly less efficient. This assumed efficiency difference also explains why the thermodynamic specific thrust prediction is so much larger. The thermodynamic specific thrust prediction is dependent on T_{t5s}, which in turn depends on the η_c. The calculated efficiency was 84% which is unrealistic (the reason for this is explained above).

The model most accurately predicted the specific fuel consumption measured by the direct assessment (error is 1.71%). Being that our estimated specific thrust was higher than the specific thrust from the direct assessment, the fuel mass flow of the model must also be higher than the fuel mass flow measured in the data.